

Sprint 2 - Farmer Marketplace Microservices Plan

Goal: Build Product Service and Inventory Service using Spring Boot Microservices while heavily practicing Stream API, Functional Interfaces, Optional, Records, Collectors, Comparator, and modern Java features.

Architecture

API Gateway -> Product Service <-> Inventory Service (Feign Client)

Product Service maintains crop catalog. Inventory Service maintains farmer-specific stock and pricing.

Epic 1 - Product Service

US-1 Create Product

Admin can create crops such as Banana, Mango, Potato, Tomato.

Tasks: Product Entity, DTOs, Repository, Service, Controller, Create API.

US-2 Get All Products

Use Stream API `map()` and method references for DTO conversion.

US-3 Search Product

Search by crop name.

US-4 Filter Products

Filter by category.

US-5 Sort Products

Sort by name and creation date.

Epic 2 - Inventory Service

US-6 Farmer Adds Inventory

Farmer publishes crop stock with quantity, price and location.

US-7 Update Inventory

Update quantity and price.

US-8 Get Inventory By Product

Buyer views all farmers selling a crop.

Epic 3 - Service Communication

US-9 Product Validation

Inventory Service validates Product existence using Feign Client.

Epic 4 - Stream API Analytics

US-10 Total Stock Per Product

US-11 Lowest Price Farmer

US-12 Highest Stock Farmer

US-13 Low Stock Alert

US-14 Inventory Valuation

Modern Java Features

Streams: filter, map, flatMap, sorted, distinct, limit, skip, peek

Collectors: groupingBy, mapping, counting, summingInt, averagingDouble

Functional Interfaces: Predicate, Function, Consumer, Supplier

Optional, Records, Switch Expressions, Comparator